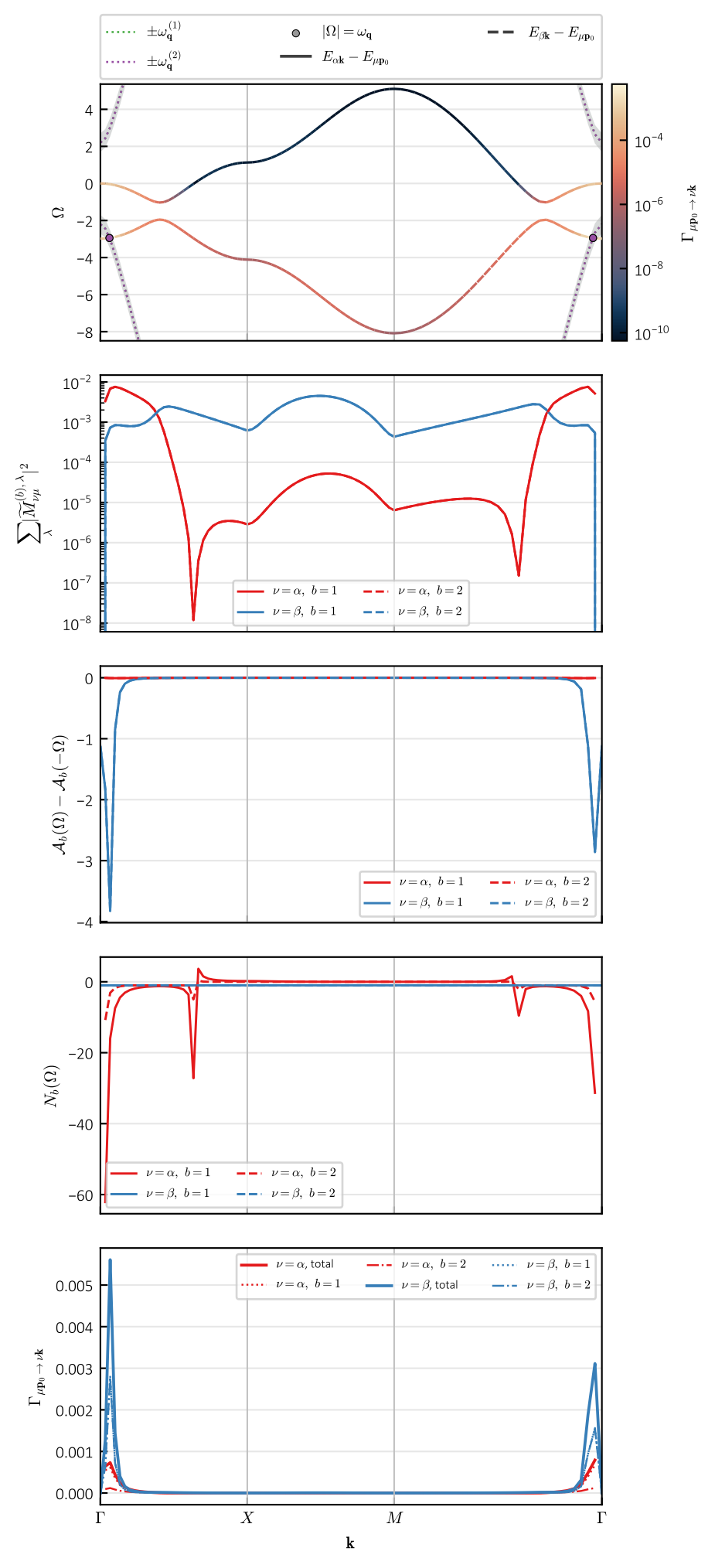
$\pm\omega_{\mathbf{q}}^{(1)}$
$\pm\omega_{\mathbf{q}}^{(2)}$
$|\Omega| = \omega_{\mathbf{q}}$
$E_{\alpha\mathbf{k}} - E_{\mu\mathbf{p}_0}$
$E_{\beta\mathbf{k}} - E_{\mu\mathbf{p}_0}$
Ω
$\Gamma_{\mu\mathbf{p}_0 \to \nu\mathbf{k}}$
$\sum_\lambda |\widetilde{M}_{\nu\mu}^{(b),\lambda}|^2$
$\nu = \alpha,\ b = 1$
$\nu = \alpha,\ b = 2$
$\nu = \beta,\ b = 1$
$\nu = \beta,\ b = 2$
$\mathcal{A}_b(\Omega) - \mathcal{A}_b(-\Omega)$
$N_b(\Omega)$
$\nu = \alpha$, total
$\nu = \beta$, total
Γ
X
M
Γ
$\mathbf{k}$